

Electromagnetic junction conditions in general relativity

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Abstract

This paper derives the electromagnetic junction conditions in general relativity when two spacetimes are joined across a non-null hypersurface of constant signature. Using the distributional approach, we determine the necessary conditions for obtaining a well-defined distributional solution to Maxwell's equations. The electromagnetic junction conditions are formulated in terms of intrinsic coordinates of the hypersurface, providing a coordinate-independent description. These conditions establish constraints on the continuity of the vector potential and the electromagnetic field tensor across the hypersurface, as well as the role of surface currents. To illustrate their application, we examine a spherically symmetric star with a surface charge density, where the exterior spacetime is Reissner-Nordström. By enforcing both the electromagnetic and Israel junction conditions, we analyze the effects of charge on the star's gravitational and electromagnetic properties. Our results highlight the interplay between charge distribution and spacetime curvature, demonstrating that surface charge contributes to the total mass through the junction conditions.

1 Introduction

In general relativity, junction conditions are boundary conditions at spacetime boundaries that arise from joining two distinct spacetimes. These spacetime boundaries are described by a hypersurface Σ that, in the most general cases, has different signatures at different points [1]. The most studied junction conditions are those of the metric and its derivatives, as the metric defines the spacetime geometry. For a simple introduction to the metric junction conditions in general relativity, see [2].

While the metric has rightfully been the primary focus of studies on junction conditions in general relativity, there are other areas of research. Examples include studies of scalar-tensor theories in modified gravity where the scalar field junction conditions must be considered alongside the metric junction conditions [3], and gravitoelectromagnetic junction conditions [4]. The metric junction conditions are also used in brane world cosmology, where our universe is a hypersurface (brane) in a higher dimensional spacetime (bulk) [5]. In these theories, gravity in the bulk can induce fields such as an electromagnetic field in the brane [6].

In this paper, we consider electromagnetic junction conditions in the context of general relativity. We derive these junction conditions using distributional methods, which are commonly used in modern derivations of the Israel junction conditions [7]. Like the Israel junction conditions, our electromagnetic junction conditions will also be limited to hypersurfaces that are either spacelike or timelike.

There are few relevant papers in the literature that explore electromagnetic junction conditions in general relativity. The most relevant one is by Rawson-Harris [8], where the covariant electromagnetic boundary conditions (junction conditions) are derived for non-null hypersurfaces. While our approach is different, we obtain similar results. Our results are also simpler

because we further reduce the junction conditions to a form that primarily depends on the intrinsic coordinates of a hypersurface.

In Section 2 we briefly introduce the Israel junction conditions. The electromagnetic junction conditions are derived in Section 3 using the Maxwell equations in curved spacetime. In Section 4 we calculate both the metric and the electromagnetic junction conditions for a static spherically symmetric star of constant density with a surface charge density such that the exterior spacetime will be Reissner-Nordström. We conclude the paper in Section 5 with a discussion of our results.

2 Geometry

In this paper, we work in $(3+1)$ -dimensional spacetime with the metric signature $(-, +, +, +)$. Let $\mathcal{V}$ be a spacetime region containing two distinct spacetime regions $\mathcal{V}^+$ and $\mathcal{V}^-$ separated by a hypersurface Σ . In $\mathcal{V}^\pm$ we have distinct coordinates $x^{\alpha\pm}$, but we assume that we can use coordinate x^α , that is distinct from $x^{\alpha\pm}$, such that it overlaps with $x^{\alpha\pm}$ in open region around Σ . We also assume that Σ is not null, since we will not be investigating the case for null Σ .

A hypersurface Σ can be described by a function

$$\Phi(x^\alpha) = 0, \quad (1)$$

or by a parametric equation

$$x^\alpha = x^\alpha(y^a), \quad (2)$$

with y^a being intrinsic coordinates of Σ [7]. (Throughout this paper, we use Latin indices to refer to the 3D coordinates of Σ , and Greek indices to refer to the 4D coordinates in $\mathcal{V}$.) Σ is characterized by its holonomic basis vectors, the normal vector, the induced metric, and the extrinsic curvature. The holonomic basis vectors are

$$e_a^\alpha = \frac{\partial x^\alpha}{\partial y^a}. \quad (3)$$

The normal vector is defined as [7]

$$n^\alpha = \frac{\varepsilon \Phi_{,\alpha}}{|g^{\mu\nu} \Phi_{,\mu} \Phi_{,\nu}|}, \quad (4)$$

where

$$n^\alpha n_\alpha = \varepsilon \equiv \begin{cases} -1 & \text{if } \Sigma \text{ is spacelike,} \\ +1 & \text{if } \Sigma \text{ is timelike.} \end{cases} \quad (5)$$

The normal vector is defined such that it points towards the direction of increasing Φ such that $n^\alpha \Phi_{,\alpha} > 0$. Usually we can figure n^α from the conditions $n^\alpha n_\alpha = \varepsilon$ and $n^\alpha u_\alpha = 0$, where u^α is four-velocity of an observer co-moving with Σ . For stationary Σ , $n^\alpha u_\alpha = 0$ is trivial, in which case $n^\alpha n_\alpha = \varepsilon$ is enough to determine n^α . The induced metric of Σ is

$$h_{ab} = g_{\alpha\beta} e_a^\alpha e_b^\beta, \quad (6)$$

and the extrinsic curvature is

$$K_{ab} = n_{\alpha;\beta} e_a^\alpha e_b^\beta. \quad (7)$$

The extrinsic curvature can also be defined in terms of the Lie derivative of the metric along the normal n :

$$K_{ab} = \frac{1}{2} (\mathcal{L}_n g_{\alpha\beta}) e_a^\alpha e_b^\beta. \quad (8)$$

The induced metric and the extrinsic curvature are important for formulating the junction conditions for the metric field in a coordinate-independent way.

2.1 Israel junction conditions

Israel junction conditions are the junction conditions for the metric field when the boundary between two spacetimes is a non-null regular hypersurface Σ . The first and the second junction conditions are [7, 9]

$$[h_{ab}] = 0, \quad [K_{ab}] = 0. \quad (9)$$

The square brackets denote the difference in quantities across Σ :

$$[A] \equiv A^+|_{\Sigma} - A^-|_{\Sigma}, \quad (10)$$

the labels for spacetime regions are such that n^α points towards $\mathcal{V}^+$.

The second junction condition $[K_{ab}] = 0$ is violated if there is a singular (thin) layer of matter at Σ . This leads to Σ having a stress tensor according to Lanczos equation:

$$S_{ab} = -\frac{\varepsilon}{8\pi}([K_{ab}] - [K]h_{ab}), \quad (11)$$

where $K = h^{ab}K_{ab}$ [2, 7]. This model with the second junction condition violated is usually called the thin shell formalism.

Now that we have a basic understanding of the geometry of hypersurfaces and the Israel junction conditions, we are ready to derive the electromagnetic junction conditions.

3 Electromagnetic junction conditions

In this section we work with the geometry of spacetime described in the previous section. We have a hypersurface Σ separating two spacetimes regions $\mathcal{V}^+$ and $\mathcal{V}^-$, where the Maxwell's equations apply. Maxwell's equations in curved spacetimes are

$$F^{\alpha\beta}_{;\beta} = 4\pi J^\alpha, \quad (12)$$

$$F_{[\alpha\beta;\gamma]} \equiv F_{\alpha\beta;\gamma} + F_{\beta\gamma;\alpha} + F_{\gamma\alpha;\beta} = 0, \quad (13)$$

where $F^{\alpha\beta}$ is the electromagnetic tensor defined via the vector potential A^μ [10]:

$$F_{\alpha\beta} = A_{\beta;\alpha} - A_{\alpha;\beta}. \quad (14)$$

We are using the convention of Misner, Thorne, and Wheeler where the electric and magnetic fields are defined by

$$F^{0j} = -F^{j0} = E^j, \quad F^{jk} = \epsilon^{jkl} B^l, \quad (15)$$

where ϵ^{jkl} is the Levi-Civita symbol [10].

We formulate the electromagnetic junction conditions by investigating the necessary conditions for obtaining valid distributional solutions to the Maxwell's equations. For the solutions to be valid they also need to produce a distributionally valid energy-momentum tensor for electromagnetism, which is defined by [10]

$$T^{(EM)}_{\alpha\beta} = \frac{1}{4\pi} \left(F_{\alpha\mu} F_{\beta}{}^{\mu} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} g_{\alpha\beta} \right). \quad (16)$$

3.1 Derivation of electromagnetic junction conditions

We start with a distributional vector potential as the solution to the Maxwell's equations

$$A_\mu = \Theta(l)A_\mu^+ + \Theta(-l)A_\mu^- . \quad (17)$$

Using the definition for the electromagnetic tensor (14) we get

$$F_{\alpha\beta} = \Theta(l)F_{\alpha\beta}^+ + \Theta(-l)F_{\alpha\beta}^- + \varepsilon\delta(l) ([A_\beta]n_\alpha - [A_\alpha]n_\beta) . \quad (18)$$

This tensor will not produce a distributionally valid energy-momentum tensor since the energy-momentum tensor is quadratic in $F_{\alpha\beta}$, and we would get terms proportional to $\Theta(l)\delta(l)$. To produce a valid energy-momentum tensor the constraint $[A_\beta]n_\alpha - [A_\alpha]n_\beta = 0$, which is equivalent to $[A_\alpha] = 0$, is necessary. Thus, our first junction condition is

$$[A_\alpha] = 0 , \quad (19)$$

and the electromagnetic tensor becomes

$$F_{\alpha\beta} = \Theta(l)F_{\alpha\beta}^+ + \Theta(-l)F_{\alpha\beta}^- . \quad (20)$$

The covariant derivative of the electromagnetic tensor is

$$F^{\alpha\beta}_{;\beta} = \Theta(l)F^{+\alpha\beta}_{;\beta} + \Theta(-l)F^{-\alpha\beta}_{;\beta} + \varepsilon\delta(l)[F^{\alpha\beta}]n_\beta , \quad (21)$$

and substituting it into the Maxwell's equations (12) and (13) gives us

$$4\pi J^\alpha = \Theta(l)F^{+\alpha\beta}_{;\beta} + \Theta(-l)F^{-\alpha\beta}_{;\beta} + \varepsilon\delta(l)[F^{\alpha\beta}]n_\beta , \quad (22)$$

$$0 = [F_{\alpha\beta}]n_\gamma + [F_{\beta\gamma}]n_\alpha + [F_{\gamma\alpha}]n_\beta . \quad (23)$$

The first equation implies

$$J^\alpha = \Theta(l)J^{+\alpha} + \Theta(-l)J^{-\alpha} + \delta(l)I^\alpha , \quad (24)$$

where $I^\alpha = J_\Sigma^\alpha$ is the four-current density on Σ . This means that the surface current density on Σ is given by

$$\varepsilon[F^{\alpha\beta}]n_\beta = 4\pi I^\alpha . \quad (25)$$

This is our second junction condition, and it simplifies to $[F^{\alpha\beta}]n_\beta = 0$ in the absence of current densities in Σ . The third and final junction condition is equation (23). It is noteworthy that this equation is trivial when indices are not unique, meaning one only needs to look at unique indices to compute this junction condition:

$$0 = [F_{\alpha\beta}]n_\gamma + [F_{\beta\gamma}]n_\alpha + [F_{\gamma\alpha}]n_\beta , \quad \text{where } \alpha \neq \beta \neq \gamma \neq \alpha . \quad (26)$$

3.1.1 Coordinate independent formulation

We can further simplify the junction conditions by writing them in terms of coordinates intrinsic to Σ such that they would be coordinate independent with regard to the spacetimes being joined.

To get coordinate independent formulation of the first electromagnetic junction condition (19), we can separate it into two equations

$$[A_a] = 0 \quad (27)$$

$$[A_\mu]n^\mu = 0 . \quad (28)$$

But it is simpler to retain the equation (19) in its original form, so we will proceed with that. As such, the first junction condition remains as

$$[A_\mu] = 0. \quad (29)$$

As for the second electromagnetic junction condition, we can start by using the antisymmetric properties of $F^{\alpha\beta}$. When we contract the second junction condition (25) with the normal vector we get

$$\varepsilon[F^{\alpha\beta}]n_\beta n_\alpha = 4\pi(I^\alpha n_\alpha), \quad (30)$$

where $I \equiv I^\alpha n_\alpha$ is a coordinate independent scalar. If $\alpha = \beta$ then $F^{\alpha\beta}$ vanishes, so we only need to look at the case where $\alpha \neq \beta$. Using antisymmetric properties of the electromagnetic tensor on the second junction condition we get

$$-\varepsilon[F^{\beta\alpha}]n_\beta = 4\pi I^\alpha \quad (31)$$

$$\Rightarrow -\varepsilon[F^{\beta\alpha}]n_\beta n_\alpha = 4\pi(I^\alpha n_\alpha) = 4\pi I. \quad (32)$$

But the second junction condition can also be written as

$$\varepsilon[F^{\beta\alpha}]n_\alpha = 4\pi I^\beta, \quad (33)$$

which when contracting gives

$$\varepsilon[F^{\beta\alpha}]n_\alpha n_\beta = 4\pi I^\beta n_\beta = 4\pi I. \quad (34)$$

Comparing this with equation (32) we get

$$[F^{\beta\alpha}]n_\alpha n_\beta = -[F^{\beta\alpha}]n_\alpha n_\beta, \quad (35)$$

which is only true if $[F^{\alpha\beta}]n_\alpha n_\beta = 0$. Thus, our second junction condition reduces to

$$\varepsilon[F^{\alpha\beta}]n_\beta = 4\pi I^\alpha, \quad (36)$$

since we can project the non-contracted component to Σ without loss of information. This tells us that the surface current doesn't have any component along the normal direction.

For the third electromagnetic junction condition we make use of the fact that equation (23):

$$0 = [F_{\alpha\beta}]n_\gamma + [F_{\beta\gamma}]n_\alpha + [F_{\gamma\alpha}]n_\beta, \quad (37)$$

is trivially satisfied whenever any two of the indices are equal, due to the antisymmetry of $F_{\alpha\beta}$. Since the idea is to work with intrinsic coordinates of Σ , the natural basis is the one with the holonomic basis vectors of Σ along with n^α as the basis vectors. In this basis n^α has only one non-zero component. Exploiting this, we can set $n_\gamma = 1$ (since any nonzero value would give the same result), which allows us to drop two terms in equation (23). The remaining term can then be projected using the holonomic basis vectors, leading to the final junction condition:

$$[F_{ab}] = 0. \quad (38)$$

All in all, the non-trivial junction conditions for electromagnetism are

$$[A_\mu] = 0, \quad (39)$$

$$\varepsilon[F^{\alpha\beta}]n_\beta = 4\pi I^\alpha, \quad (40)$$

$$[F_{ab}] = 0. \quad (41)$$

3.2 Energy-momentum tensor

As a consequence of the first electromagnetic junction conditions, the electromagnetic energy-momentum tensor has the form

$$T^{(EM)}_{\alpha\beta} = \Theta(l)T^+_{\alpha\beta} + \Theta(-l)T^-_{\alpha\beta}, \quad (42)$$

where $T^{\pm}_{\alpha\beta} = \frac{1}{4\pi} \left(F^{\pm}_{\alpha\mu} F^{\pm\mu}_{\beta} - \frac{1}{4} F^{\pm}_{\mu\nu} F^{\pm\mu\nu} g^{\pm}_{\alpha\beta} \right)$ with the $\pm$ signs describing the quantities in different sides Σ . This result tells us that the energy-momentum tensor produced by an electromagnetic field cannot be singular even if there is charge density at Σ and the electromagnetic tensor $F_{\alpha\beta}$ is discontinuous. This is not surprising since $T^{\alpha\beta}$ is quadratic in $F_{\alpha\beta}$ —this restricts $F_{\alpha\beta}$ to be regular at Σ —but it is an interesting contrast to the energy-momentum tensor of matter, which can be singular at Σ .

4 A star with a surface charge density

As an example, we present a calculation of junction conditions for a star that has a uniform charge density at the surface while the rest of the star is electrically neutral.

Let us consider the following situation, as depicted in figure 1. We have an exterior Reissner-Nordström (RN) spacetime $\mathcal{V}^+$, joined to an interior spacetime $\mathcal{V}^-$, representing a star with constant density ρ_* and total charge Q , which is entirely confined to its surface. The hypersurface Σ separating these regions is a time-like hypersurface. The radial coordinate r is an affine parameter that is smooth across the boundary and the same on both sides. The coordinates on Σ can be straightforwardly chosen to be $y^a = (t, \theta, \phi)$.

We assume an idealized star that is static with constant density ρ_* and is spherically symmetric. The following metric in Schwarzschild coordinates describes the inner part of this star

$$ds^2 = - \left[\frac{3}{2} \left(1 - \frac{2M_\star}{R} \right)^{\frac{1}{2}} - \frac{1}{2} \left(1 - \frac{2M_\star r^2}{R^3} \right)^{\frac{1}{2}} \right]^2 dt^2 + \left(1 - \frac{2m(r)}{r} \right)^{-1} dr^2 + r^2 d\Omega^2, \quad (43)$$

where $m(r) = \frac{4\pi}{3} \rho_* r^3$ is the mass enclosed within radius r , R is the radius of the star, and $M_\star = m(R)$ is the total mass contributed by the density of the star [10]. Outside the star, in $\mathcal{V}^+$, the metric is RN

$$ds^2 = -f_{RN} dt^2 + f_{RN}^{-1} dr^2 + r^2 d\Omega^2, \quad (44)$$

with $f_{RN} = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}$, where M is the total gravitational mass distinct from $M_\star$ due to the contribution of the total charge Q to the spacetime curvature.

Let us first calculate the Israel junction conditions to see the conditions for joining the spacetimes. We must first compute the induced metric and the extrinsic curvature to calculate the Israel junction conditions. The induced metric is given by

$$h_{ab} = g_{\alpha\beta} e^\alpha_a e^\beta_b, \quad (45)$$

which means that the metrics on the two sides of Σ (at $r = R$) are:

$$ds^2|_{\Sigma^-} = - \left(1 - \frac{2M_\star}{R} \right) dt^2 + R^2 d\Omega^2, \quad (46)$$

$$ds^2|_{\Sigma^+} = -f_{RN} dt^2 + R^2 d\Omega^2. \quad (47)$$

The first Israel junction condition is $[h_{ab}] = 0$, this gives us the relation

$$f_{RN}|_{r=R} = \left(1 - \frac{2M_\star}{R} \right) \quad (48)$$

$$\Rightarrow M = M_\star + \frac{1}{2} \frac{Q^2}{R}. \quad (49)$$

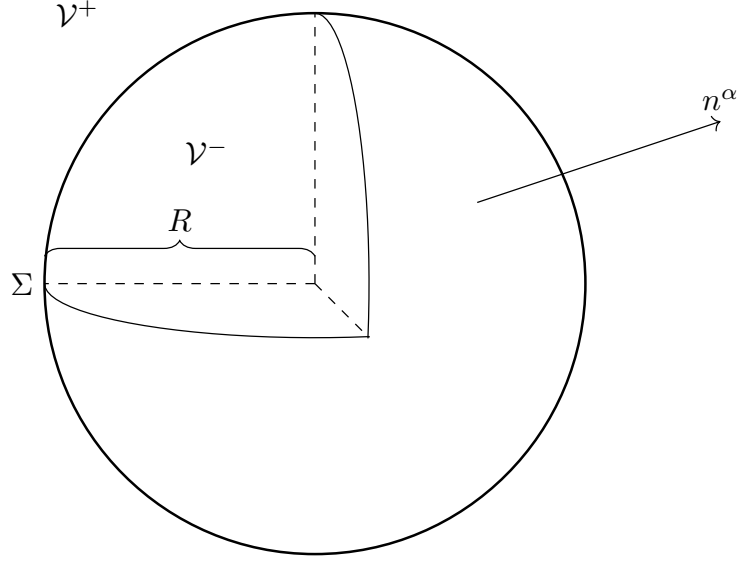


Figure 1: Visual representation of the star: manifolds $\mathcal{V}^-$ and $\mathcal{V}^+$ and the boundary Σ between them. The vector n^α is the normal vector that points to the spacetime $\mathcal{V}^+$ from the hypersurface Σ .

This relation shows that the total mass has a charge contribution in addition to the stellar mass $M_\star = \frac{4\pi}{3}\rho_\star R^3$, and this contribution is singular just like the charge distribution itself. Let's now calculate the extrinsic curvature using equation (8):

$$K_{ab} = \frac{1}{2} \mathcal{L}_n g_{\mu\nu} e_a^\mu e_b^\nu \quad (50)$$

$$= \frac{1}{2} (n^\alpha \partial_\alpha g_{\mu\nu} + (\partial_\mu n^\alpha) g_{\nu\alpha} + (\partial_\nu n^\alpha) g_{\mu\alpha}) e_a^\mu e_b^\nu. \quad (51)$$

To determine n^α , we use the conditions $n^\alpha n_\alpha = 1$ and $n^\alpha u_\alpha = 0$, where u^α is the four-velocity of Σ . These conditions give us $n^r = \frac{1}{\sqrt{g_{rr}}}$ as the only non-zero component. Thus, equation (51) simplifies to

$$K_{ab} = \frac{1}{2} (n^r \partial_r g_{\mu\nu} + (\partial_\mu n^r) g_{\nu r} + (\partial_\nu n^r) g_{\mu r}) e_a^\mu e_b^\nu. \quad (52)$$

Since the metric for both of our spacetimes are diagonal, Σ is at constant r , and its coordinates are $y^a = (t, \theta, \phi)$, we have $e_a^r = 0$ which means the expression above simplifies to

$$K_{ab} = \frac{1}{2} (n^r \partial_r g_{\mu\nu}) e_a^\mu e_b^\nu.$$

Calculating the extrinsic curvatures on both sides of Σ gives us the following results:

$$K_{tt}^- = -\frac{M_\star}{R^2 \sqrt{g_{RR}^-}} \quad K_{tt}^+ = \frac{1}{R^2 \sqrt{g_{RR}^+}} \left(-M + \frac{Q^2}{R} \right) \quad (53)$$

$$K_{\theta\theta}^- = \frac{R}{\sqrt{g_{RR}^-}} \quad K_{\theta\theta}^+ = \frac{R}{\sqrt{g_{RR}^+}} \quad (54)$$

$$K_{\phi\phi}^- = \frac{R \sin^2 \theta}{\sqrt{g_{RR}^-}} \quad K_{\phi\phi}^+ = \frac{R \sin^2 \theta}{\sqrt{g_{RR}^+}}, \quad (55)$$

where $g_{RR}^\pm \equiv g_{rr}^\pm|_{r=R}$. We see from equations (53)–(55) that $[K_{\theta\theta}] = [K_{\phi\phi}] = 0$, due to the first junction condition. The only component with possible discontinuity is K_{tt} :

$$[K_{tt}] = \frac{1}{R^2 \sqrt{g_{RR}}} \left(-M + M_\star + \frac{Q^2}{R} \right), \quad (56)$$

where $\sqrt{g_{RR}} = \sqrt{g_{RR}^+} = \sqrt{g_{RR}^-}$. Using the relation (49) from the first junction condition, we get

$$[K_{tt}] = \frac{1}{2R^2 \sqrt{g_{RR}}} \left(\frac{Q^2}{R} \right). \quad (57)$$

Thus the second junction condition is violated and there is a thin shell at Σ . To see what contributes to this thin shell, let us calculate the energy-momentum tensor of Σ using the Lanczos equation (11)

$$S_{ab} = -\frac{\varepsilon}{8\pi} ([K_{ab}] - [K]h_{ab}), \quad (58)$$

where $K = h^{ab}K_{ab}$. Using second junction condition we get $[K] = [h^{tt}K_{tt}] = g^{tt}[K_{tt}]$, where $g^{tt} = (g_{tt})^{-1}$. This simplifies the Lanczos equation (58) to

$$S_{ab} = -\frac{\varepsilon}{8\pi} [K_{tt}] (\delta_{at}\delta_{bt} - (g_{tt})^{-1}h_{ab}). \quad (59)$$

Using the second junction condition (57), we get the components of S_{ab} :

$$S_{tt} = 0 \quad (60)$$

$$S_{\theta\theta} = -\frac{1}{16\pi} \sqrt{g_{RR}} \left(\frac{Q^2}{R} \right) \quad (61)$$

$$S_{\theta\theta} = -\frac{1}{16\pi} \sqrt{g_{RR}} \left(\frac{Q^2}{R} \right) \sin^2 \theta, \quad (62)$$

where we used $g_{RR}^\pm = -(g_{tt}^\pm)^{-1}|_{r=R}$ and $\varepsilon = 1$.

The singular charge density produces a singular energy-momentum tensor at Σ , not because of the mass of the charge carriers, which is included in the mass $M_\star$, but from the charge's contribution to the total mass of the system via equation (49). This equation tells that there is an extra layer effective of mass $M_\Sigma = \frac{1}{2} \frac{Q^2}{R}$ at Σ . This is also evident from the Q^2 dependence of the singular part of the energy-momentum tensor. It is an interesting result that while the electromagnetic fields themselves cannot contribute to the singular part of the energy-momentum tensor, the charge density does so via its effect on the spacetime curvature.

Let us now use the previously derived electromagnetic junction conditions. In $\mathcal{V}^+$ we have the stress tensor $F^{\alpha\beta}$ with the non-zero components $F^{tr} = \frac{Q}{r^2} = -F^{rt}$. In $\mathcal{V}^-$, where there is no electromagnetic field we have $F^{\alpha\beta} = 0$. The first electromagnetic junction condition is trivial because we can set the potential in V^- to be constant such that $A_\alpha^-(r) = A_\alpha^+(R)$. The third electromagnetic junction condition is also trivial since the only non-zero components of the electromagnetic tensor in $\mathcal{V}^+$ are $F^{tr} = -F^{rt}$ and r is the index along the normal, which means $F_{ab}^+ = 0 \Rightarrow [F_{ab}] = 0$. The second electromagnetic junction condition is not trivial since we have a star with a surface charge density:

$$\varepsilon[F^{a\beta}]n_\beta = 4\pi I^a \quad (63)$$

$$\Rightarrow \frac{Q}{R^2} \sqrt{g_{RR}} = 4\pi I^t. \quad (64)$$

This means that the charge density observed by a stationary observer, whose four-velocity is $u_\alpha = \sqrt{-g_{tt}}\delta_{t\alpha}$, is

$$I^t u_t = \frac{Q}{4\pi R^2} \sqrt{g_{RR}} \sqrt{-g_{tt}} = \frac{Q}{4\pi R^2}. \quad (65)$$

This is exactly the surface charge density we had assigned the surface. Thus, a star with a surface charge density satisfies the electromagnetic junction conditions.

5 Discussion

We have derived electromagnetic junction conditions for timelike and spacelike hypersurfaces using distributional methods. Except for the first one, these junction conditions only depend on the intrinsic coordinates of the hypersurface. Our coordinate-independent formulation is new compared to the derivation of electromagnetic junction conditions by Rawson-Harris [8]. The advantage of Rawson-Harris's approach is that it works with hypersurfaces that can arbitrarily change signature between timelike and spacelike. The advantage of our approach, in addition to simpler equations, is that it does not require the electromagnetic sources to be independent of time and works for dynamic situations.

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